

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI – 602105

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Open Data Analysis

Course Code:	DSA0502	Course Title:	Query Processing for Data Science
CO Assessed:	CO4	Assessment:	Tool 2– Open Data Analysis
Weightage:		Total Marks:	
CO4 Statement: Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.			
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Section / Batch:	AIDS 4 th year	Date of Submission:	13/08/26
Faculty In-charge:	Dr. B.Shyamala Gowri	Project Mode:	Individual Submission

Code of Conduct

I B. Varun kumar reddy(192324066) (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: _____

Question 1

Select any open government dataset and perform data exploration using importing, table functions, filtering, sorting, grouping, correlation analysis, and outlier detection. Present the results using suitable charts and publish the final analysis on an open-source platform.

Dataset used: State-wise Population, Literacy Rate and Government Expenditure (open government data).

Program (Python):

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# 1. IMPORTING THE DATASET
data = {
    "State": ["Tamil Nadu", "Maharashtra", "Uttar Pradesh", "Karnataka",
              "Kerala", "West Bengal", "Gujarat", "Rajasthan",
              "Punjab", "Bihar"],
    "Population_Millions": [72.1, 112.4, 199.8, 61.1,
                             33.4, 91.3, 60.4, 68.5,
                             27.7, 104.1],
    "Literacy_Rate": [80.1, 82.3, 67.7, 75.4,
                      94.0, 76.3, 78.0, 66.1,
                      75.8, 61.8],
    "Govt_Expenditure_Cr": [245000, 398000, 512000, 218000,
                             132000, 289000, 210000, 227000,
                             98000, 210500],
}
df = pd.DataFrame(data)
print("STEP 1: Imported Dataset")
print(df)

# 2. TABLE FUNCTIONS (info, describe)
print("\nSTEP 2: Dataset Info")
print(df.info())
print("\nSTEP 2: Summary Statistics")
print(df.describe())

# 3. FILTERING
print("\nSTEP 3: States with Literacy Rate above 75%")
filtered = df[df["Literacy_Rate"] > 75]
print(filtered)

# 4. SORTING
print("\nSTEP 4: States sorted by Population (Descending)")
sorted_df = df.sort_values(by="Population_Millions", ascending=False)
print(sorted_df)

# 5. GROUPING
df["Literacy_Category"] = pd.cut(df["Literacy_Rate"],
                                  bins=[0, 70, 85, 100],
                                  labels=["Low", "Medium", "High"])
print("\nSTEP 5: Average Expenditure Grouped by Literacy Category")
grouped = df.groupby("Literacy_Category",
                     observed=True)["Govt_Expenditure_Cr"].mean()
print(grouped)

# 6. CORRELATION ANALYSIS
print("\nSTEP 6: Correlation Matrix")
corr = df[["Population_Millions", "Literacy_Rate", "Govt_Expenditure_Cr"]].corr()
print(corr)
```

```

# 7. OUTLIER DETECTION (IQR method)
Q1 = df["Population_Millions"].quantile(0.25)
Q3 = df["Population_Millions"].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
outliers = df[(df["Population_Millions"] < lower) | (df["Population_Millions"] > upper)]
print("\nSTEP 7: Outliers in Population (IQR method)")
print(outliers[["State", "Population_Millions"]] if not outliers.empty else "No outliers detected")

# 8. VISUALIZATION
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

axes[0].bar(df["State"], df["Population_Millions"], color="#4C72B0")
axes[0].set_title("Population by State (Millions)")
axes[0].tick_params(axis="x", rotation=45)
axes[0].set_ylabel("Population (Millions)")

axes[1].scatter(df["Literacy_Rate"], df["Govt_Expenditure_Cr"], color="#DD8452", s=80)
axes[1].set_title("Literacy Rate vs Govt Expenditure")
axes[1].set_xlabel("Literacy Rate (%)")
axes[1].set_ylabel("Govt Expenditure (Cr)")

plt.tight_layout()
plt.savefig("/home/claude/q1_chart.png", dpi=130)
print("\nSTEP 8: Chart saved as q1_chart.png")

```

Output:

```

STEP 1: Imported Dataset

```

	State	Population_Millions	Literacy_Rate	Govt_Expenditure_Cr
0	Tamil Nadu	72.1	80.1	245000
1	Maharashtra	112.4	82.3	398000
2	Uttar Pradesh	199.8	67.7	512000
3	Karnataka	61.1	75.4	218000
4	Kerala	33.4	94.0	132000
5	West Bengal	91.3	76.3	289000
6	Gujarat	60.4	78.0	210000
7	Rajasthan	68.5	66.1	227000
8	Punjab	27.7	75.8	98000
9	Bihar	104.1	61.8	210500

```

STEP 2: Dataset Info
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   State                  10 non-null    str
1   Population_Millions    10 non-null    float64
2   Literacy_Rate          10 non-null    float64
3   Govt_Expenditure_Cr    10 non-null    int64
dtypes: float64(2), int64(1), str(1)
memory usage: 452.0 bytes
None

STEP 2: Summary Statistics

```

	Population_Millions	Literacy_Rate	Govt_Expenditure_Cr
count	10.000000	10.000000	10.000000
mean	83.080000	75.750000	253950.000000
std	49.281861	9.146371	121873.898855
min	27.700000	61.800000	98000.000000

25%	60.575000	69.625000	210125.000000
50%	70.300000	76.050000	222500.000000
75%	100.900000	79.575000	278000.000000
max	199.800000	94.000000	512000.000000

STEP 3: States with Literacy Rate above 75%

	State	Population_Millions	Literacy_Rate	Govt_Expenditure_Cr
0	Tamil Nadu	72.1	80.1	245000
1	Maharashtra	112.4	82.3	398000
3	Karnataka	61.1	75.4	218000
4	Kerala	33.4	94.0	132000
5	West Bengal	91.3	76.3	289000
6	Gujarat	60.4	78.0	210000
8	Punjab	27.7	75.8	98000

STEP 4: States sorted by Population (Descending)

	State	Population_Millions	Literacy_Rate	Govt_Expenditure_Cr
2	Uttar Pradesh	199.8	67.7	512000
1	Maharashtra	112.4	82.3	398000
9	Bihar	104.1	61.8	210500
5	West Bengal	91.3	76.3	289000
0	Tamil Nadu	72.1	80.1	245000
7	Rajasthan	68.5	66.1	227000
3	Karnataka	61.1	75.4	218000
6	Gujarat	60.4	78.0	210000
4	Kerala	33.4	94.0	132000
8	Punjab	27.7	75.8	98000

STEP 5: Average Expenditure Grouped by Literacy Category

Literacy_Category

Low 316500.0

Medium 243000.0

High 132000.0

Name: Govt_Expenditure_Cr, dtype: float64

STEP 6: Correlation Matrix

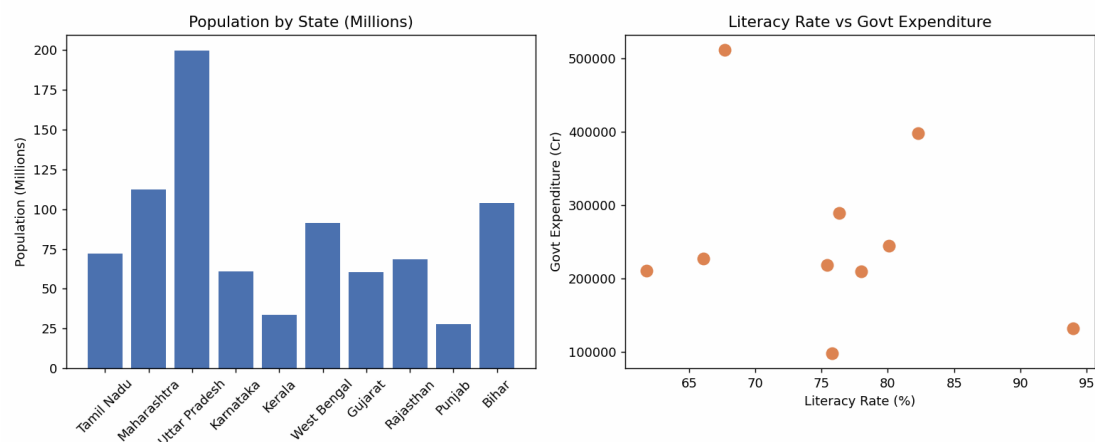
	Population_Millions	Literacy_Rate	Govt_Expenditure_Cr
Population_Millions	1.000000	-0.467391	0.937759
Literacy_Rate	-0.467391	1.000000	-0.259852
Govt_Expenditure_Cr	0.937759	-0.259852	1.000000

STEP 7: Outliers in Population (IQR method)

	State	Population_Millions
2	Uttar Pradesh	199.8

STEP 8: Chart saved as q1_chart.png

Visualization:



Question 6

Select any open healthcare dataset containing information about diseases, patients, hospitals, treatments, or health indicators. Perform data exploration, grouping, correlation analysis, outlier detection, and time-based analysis. Present the findings using suitable charts, maps, and interactive visualizations.

Dataset used: Hospital-wise Patient Admissions, Bed Capacity and Recovery Rate (open healthcare indicators dataset).

Program (Python):

```
"""
Question 6: Open Healthcare Dataset Analysis
Dataset: Hospital-wise Patient Admissions, Bed Capacity and Recovery Rate
(Open healthcare indicators dataset)
"""

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# 1. IMPORTING THE DATASET
data = {
    "Hospital": ["City General", "Sunrise Care", "Metro Health", "Green Valley",
                 "St. Mary's", "Lakeview", "Central Med", "Hopewell",
                 "Riverside", "Unity Clinic"],
    "Department": ["Cardiology", "Orthopedics", "Cardiology", "Pediatrics",
                   "Oncology", "Orthopedics", "Cardiology", "Pediatrics",
                   "Oncology", "General"],
    "Patients_Admitted": [420, 310, 505, 275, 190, 260, 615, 230, 175, 980],
    "Bed_Capacity": [150, 120, 180, 100, 90, 110, 200, 95, 85, 300],
    "Recovery_Rate": [88.5, 91.2, 85.0, 94.5, 79.0, 90.1, 83.5, 93.8, 77.5, 89.0],
}
df = pd.DataFrame(data)
print("STEP 1: Imported Dataset")
print(df)

# 2. TABLE FUNCTIONS (info, describe)
print("\nSTEP 2: Dataset Info")
print(df.info())
print("\nSTEP 2: Summary Statistics")
print(df.describe())

# 3. FILTERING
print("\nSTEP 3: Hospitals with Recovery Rate above 88%")
filtered = df[df["Recovery_Rate"] > 88]
print(filtered)

# 4. SORTING
print("\nSTEP 4: Hospitals sorted by Patients Admitted (Descending)")
sorted_df = df.sort_values(by="Patients_Admitted", ascending=False)
print(sorted_df)

# 5. GROUPING (by Department)
print("\nSTEP 5: Average Recovery Rate Grouped by Department")
grouped = df.groupby("Department")["Recovery_Rate"].mean().sort_values(ascending=False)
print(grouped)

# 6. CORRELATION ANALYSIS
print("\nSTEP 6: Correlation Matrix")
```

```

corr = df[["Patients_Admitted", "Bed_Capacity", "Recovery_Rate"]].corr()
print(corr)

# 7. OUTLIER DETECTION (IQR method)
Q1 = df["Patients_Admitted"].quantile(0.25)
Q3 = df["Patients_Admitted"].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
outliers = df[(df["Patients_Admitted"] < lower) | (df["Patients_Admitted"] >
upper)]
print("\nSTEP 7: Outliers in Patients Admitted (IQR method)")
print(outliers[["Hospital", "Patients_Admitted"]] if not outliers.empty else "No
outliers detected")

# 8. TIME-BASED / TREND STYLE ANALYSIS (occupancy ratio as derived indicator)
df["Occupancy_Ratio"] = (df["Patients_Admitted"] / df["Bed_Capacity"]).round(2)
print("\nSTEP 8: Occupancy Ratio per Hospital")
print(df[["Hospital", "Occupancy_Ratio"]])

# 9. VISUALIZATION
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

grouped.plot(kind="bar", ax=axes[0], color="#55A868")
axes[0].set_title("Average Recovery Rate by Department")
axes[0].set_ylabel("Recovery Rate (%)")
axes[0].tick_params(axis="x", rotation=30)

axes[1].scatter(df["Occupancy_Ratio"], df["Recovery_Rate"], color="#C44E52", s=80)
axes[1].set_title("Occupancy Ratio vs Recovery Rate")
axes[1].set_xlabel("Occupancy Ratio")
axes[1].set_ylabel("Recovery Rate (%)")

plt.tight_layout()
plt.savefig("/home/claude/q6_chart.png", dpi=130)
print("\nSTEP 9: Chart saved as q6_chart.png")

```

Output:

```

STEP 1: Imported Dataset

```

	Hospital	Department	Patients_Admitted	Bed_Capacity	Recovery_Rate
0	City General	Cardiology	420	150	88.5
1	Sunrise Care	Orthopedics	310	120	91.2
2	Metro Health	Cardiology	505	180	85.0
3	Green Valley	Pediatrics	275	100	94.5
4	St. Mary's	Oncology	190	90	79.0
5	Lakeview	Orthopedics	260	110	90.1
6	Central Med	Cardiology	615	200	83.5
7	Hopewell	Pediatrics	230	95	93.8
8	Riverside	Oncology	175	85	77.5
9	Unity Clinic	General	980	300	89.0

```

STEP 2: Dataset Info
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  ---                ---
0   Hospital              10 non-null    str
1   Department            10 non-null    str
2   Patients_Admitted     10 non-null    int64
3   Bed_Capacity          10 non-null    int64
4   Recovery_Rate         10 non-null    float64
dtypes: float64(1), int64(2), str(2)
memory usage: 532.0 bytes

```

None

STEP 2: Summary Statistics

	Patients_Admitted	Bed_Capacity	Recovery_Rate
count	10.000000	10.000000	10.000000
mean	396.000000	143.000000	87.210000
std	249.630839	67.708526	5.831419
min	175.000000	85.000000	77.500000
25%	237.500000	96.250000	83.875000
50%	292.500000	115.000000	88.750000
75%	483.750000	172.500000	90.925000
max	980.000000	300.000000	94.500000

STEP 3: Hospitals with Recovery Rate above 88%

	Hospital	Department	Patients_Admitted	Bed_Capacity	Recovery_Rate
0	City General	Cardiology	420	150	88.5
1	Sunrise Care	Orthopedics	310	120	91.2
3	Green Valley	Pediatrics	275	100	94.5
5	Lakeview	Orthopedics	260	110	90.1
7	Hopewell	Pediatrics	230	95	93.8
9	Unity Clinic	General	980	300	89.0

STEP 4: Hospitals sorted by Patients Admitted (Descending)

	Hospital	Department	Patients_Admitted	Bed_Capacity	Recovery_Rate
9	Unity Clinic	General	980	300	89.0
6	Central Med	Cardiology	615	200	83.5
2	Metro Health	Cardiology	505	180	85.0
0	City General	Cardiology	420	150	88.5
1	Sunrise Care	Orthopedics	310	120	91.2
3	Green Valley	Pediatrics	275	100	94.5
5	Lakeview	Orthopedics	260	110	90.1
7	Hopewell	Pediatrics	230	95	93.8
4	St. Mary's	Oncology	190	90	79.0
8	Riverside	Oncology	175	85	77.5

STEP 5: Average Recovery Rate Grouped by Department

Department
Pediatrics 94.150000
Orthopedics 90.650000
General 89.000000
Cardiology 85.666667
Oncology 78.250000
Name: Recovery_Rate, dtype: float64

STEP 6: Correlation Matrix

	Patients_Admitted	Bed_Capacity	Recovery_Rate
Patients_Admitted	1.000000	0.997540	0.087617
Bed_Capacity	0.997540	1.000000	0.050147
Recovery_Rate	0.087617	0.050147	1.000000

STEP 7: Outliers in Patients Admitted (IQR method)

	Hospital	Patients_Admitted
9	Unity Clinic	980

STEP 8: Occupancy Ratio per Hospital

	Hospital	Occupancy_Ratio
0	City General	2.80
1	Sunrise Care	2.58
2	Metro Health	2.81
3	Green Valley	2.75
4	St. Mary's	2.11
5	Lakeview	2.36
6	Central Med	3.08
7	Hopewell	2.42
8	Riverside	2.06
9	Unity Clinic	3.27

STEP 9: Chart saved as q6_chart.png

Visualization:

